

BERNARDO BARRERA

bbarrera@bu.edu

EDUCATION

Ph.D. in Physics

Boston University

Supervised by Anushya Chandran and Anatoli Polkovnikov

August 2022–June 2028 (Expected)

B.S. in Engineering Physics

Instituto Tecnológico y de Estudios Superiores de Monterrey

August 2017–June 2022

RESEARCH EXPERIENCE

Non-Adiabatic Dynamics and Mixed Quantum-Classical Methods

BU Condensed Matter Theory Group

January 2023–Present

- Developed a mixed quantum-classical framework extending the Born-Oppenheimer approximation beyond the adiabatic limit, published in the *Proceedings of the National Academy of Sciences*.
- Contributed core ideas, analytical derivations, numerical algorithms and simulations.

Topology in Circuit QED

In collaboration with the Kollár Group at UMD

May 2025–Present

- Theory student in a project developing the first experimental realization of a quantum topological photon pump using a transmon qubit coupled to a microwave cavity.
- Contributed theoretical models and numerical simulations to understand experimental data and guide new experimental directions.

Dynamics in Classical and Quantum Billiards

Photonics and Mathematical Optics Group, Tecnológico de Monterrey

January 2022–August 2022

- Studied the classical and quantum dynamics of an elliptic billiard in a harmonic potential, published in *Physical Review E*.

Gravitational Lensing Models

University of Minnesota

June 2020–August 2020

- Developed models for the mass distribution in the lensing galaxy of a gravitationally lensed quasar system, published in *The Open Journal of Astrophysics*.

PUBLICATIONS

Barrera, B., Verma, N., Queiroz, R., Polkovnikov, A., Chandran, A. (2026). Kinematic Quantum Geometry. *Manuscript in preparation*.

Yue, Q., **Barrera, B.**, Ritter, M., Long, D. M., Lane, D. A., Chandran, A., & Kollár, A. J. (2026). Realization of a quantum topological photon pump. *arXiv:2608.00162*.

Barrera, B., Arovas, D. P., Chandran, A., & Polkovnikov, A. (2026). The moving Born–Oppenheimer approximation. *Proceedings of the National Academy of Sciences*, 123(7), e2507816123.

Barrera, B., Ruz-Cuen, J. P., & Gutiérrez-Vega, J. C. (2023). Elliptic billiard with harmonic potential: Classical description. *Physical Review E*, 108(3), 034205.

Barrera, B., Williams, L. L. R., Coles, J. P., & Denzel, P. (2021). Bridging the gap between simply parametrized and free-form pixelated models of galaxy lenses: The case of WFI 2033-4723 Quad. *The Open Journal of Astrophysics*, 4(1).

AWARDS

Goldhaber Prize for early excellence in first year of graduate school <i>Boston University</i>	<i>2024</i>
Teaching Fellow of the Year <i>Boston University</i>	<i>2023</i>
Best GPA of the School of Engineering and Sciences <i>Instituto Tecnológico y de Estudios Superiores de Monterrey</i>	<i>2022</i>
1st Place Mexican ANFEI National Science Competition <i>Instituto Tecnológico y de Estudios Superiores de Monterrey</i>	<i>2019</i>
Academic Excellence Scholarship <i>Instituto Tecnológico y de Estudios Superiores de Monterrey</i>	<i>2017–2022</i>

TALKS AND POSTERS

- Poster at the 2026 Molecular Interactions and Dynamics Gordon Research Conference.
Title: *Geometric Phases and Forces in Chiral Pseudospin Textures*.
- Contributed talk at the 2026 APS Global Physics Summit.
Title: *Thermodynamics of Mixed Quantum-Classical Systems*.
- Invited talk at ETH Zurich ITP, 2025.
Title: *The Moving Born–Oppenheimer Approximation*.
- Invited talk at the 2025 Chaos and Geometry in Quantum Many-Body Systems Conference, Ettore Majorana Foundation and Centre for Scientific Culture.
Title: *The Moving Born–Oppenheimer Approximation*.
- Contributed talk at the 2025 APS Global Physics Summit.
Title: *Non-Adiabatic Dynamics Beyond the Born–Oppenheimer Approximation*.
- Contributed talk at the 2024 APS March Meeting.
Title: *Non-Adiabatic Dynamics Beyond the Born–Oppenheimer Approximation*.

TEACHING

- Teaching assistant for PY211 General Physics I at Boston University, Fall 2022 and 2023.
- Teaching assistant for PY212 General Physics II at Boston University, Spring 2023 and 2024.
- Teaching assistant for PY351 Modern Physics at Boston University, Summer 2023.

SKILLS

Physics	Quantum many-body dynamics and simulation, phase space representations of quantum dynamics, mixed quantum-classical algorithms, non-adiabatic dynamics, quantum geometry and topology, circuit QED, superconducting qubits
Programming Languages	Julia, Python, MATLAB Spanish (native), English (fluent), Portuguese (intermediate)